

This version adds three lines under the title (a pointer to One coordinate up, where the corpus is now read) and carries one reread and one clarification; the identity, every threshold and every value stand.

- (1) Three lines added under the title – READING UPDATE – pointing to One coordinate up (<https://osf.io/ujwc7> · DOI 10.17605/OSF.IO/UJWC7), the synthesis where the corpus is now read.
- (2) This paper is classed as a NOTE on the Observational Record shelf of the framework's Direct Documents page, under the short title "The Hodgson–Kerckhoff Closed Form".
- (3) REREAD, Section 7, item 4: "DESI BAO: residual at 50th data point matches framework prediction" misstated the computation. There is no 50th data point – DESI DR2 Table IV has seven redshift bins; the computed fact is the residual at the $1/50^2$ scale: $DV(LRG2)/DV(BGS) - (2 + G) = +0.000399$, against $1/2500 = 0.000400$ (The 50 Hinge, §7, from Table IV). The item is reread in place to say that; the four contexts of the 50 Hinge stand.
- (4) PRINTED-DIGIT CORRECTION, Section 9 (verification code): the comment "# = 3.39571381800..." for $A = 2\sqrt{6} \cdot \ln 2$ corrected to 3.39571381804... (exact 3.3957138180413...). Caught by T00LBOX/printed_digits.py on the reprint; the code computes A correctly, the comment misprinted it.
- (5) CLARIFICATION, Section 8: "0.980258 (numerical, no closed form until this paper)" – Hodgson and Kerckhoff write S in closed hyperbolic form, $S = (1/(2\sqrt{2})) / \operatorname{arcsinh}(1/(2\sqrt{2}))$, as Section 2 of this paper quotes; what is new here is the logarithmic simplification $\operatorname{arcsinh}(1/(2\sqrt{2})) = \ln(2)/2$. A bracket in place says so.

Every value re-computed 2026-09-11 at 40 digits (the identity, R_0 and its five hyperbolic values, $A = 2\sqrt{6} \cdot \ln 2$, the three thresholds and their spacings, the 0.51 % and 1.26 % landmarks, the Ramanujan–Nagell set, 181 as the 42nd prime): all stand. Prior version remains on OSF in the project's archive/ folder, named by its date.

This version corrects the Landauer-crossing framing in Sections 4 and 12. The earlier text placed the crossing "between step 36 and step 37." The refined picture identifies THREE distinct thresholds that cluster in the 35–37 zone:

$n \cdot G = \ln 2$	at $n \approx 35.11$	(Landauer erasure cost – the crossing)
$n \cdot G = 1/\sqrt{2}$	at $n \approx 35.82$	(geometric damping)
$n \cdot G = 1/(2 \ln 2)$	at $n \approx 36.54$	(Shannon bound)

The Landauer crossing proper is the FIRST threshold, at $n = \ln(2)/G$, which sits between integer steps 35 and 36. Integer steps 36 and 37 are landmarks NEAR the second and third thresholds, not the crossing itself.

The exact algebraic identity $1/(2 \ln 2) - 1/\sqrt{2} = G/(2 \ln 2)$ is unchanged. It is renamed from "Landauer crossing identity" to " $\sqrt{2}$ -to-Shannon gap identity" because it describes the spacing between thresholds 2 and 3 (geometric damping → Shannon bound), not the Landauer crossing itself.

Section 4 is expanded with a three-threshold breakdown. Section 12 is

reworded to reflect that the crossing zone unfolds across three thresholds, with the spacings algebraically locked: $\Delta n_{12} = 1/\sqrt{2}$, $\Delta n_{23} = 1/(2 \ln 2)$, $\Delta n_{23}/\Delta n_{12} = 1/K_{\text{AUD}}$.

Prior published versions remain on OSF file storage, renamed with the OLD- prefix and a date for traceability.

NOTE · The Hodgson–Kerckhoff Closed Form

THE HODGSON-KERCKHOFF CLOSED FORM
AND THE K_{AUD} FRAMEWORK

A Level 1 Algebraic Identity in Published Hyperbolic Geometry

D. B. – Gap Geometry Project – April 2026
(Updated 13 April 2026; 16 September 2026)

■ READING UPDATE – 2026-09-16. No value in this paper is changed by this notice.
How the corpus is now read: One coordinate up – <https://osf.io/ujwc7>
DOI 10.17605/OSF.IO/UJWC7

Gap Geometry – <https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>
OSF: osf.io/zx4g7
Cross-architecture verification: Grok (xAI) + Claude Opus (Anthropic)

ABSTRACT

The constant $K_{\text{AUD}} = \sqrt{2} * \ln(2) = 0.980258143\dots$ is the unique sub-unity value of $K(n) = \sqrt{n} * \ln(n)$ for integer $n \geq 2$. It defines an irreducible gap $G = 1 - K_{\text{AUD}} \sim 0.01974$ between binary information density and unity.

This paper reports that the numerical coefficient 0.980258 appearing in the tube-packing bound of Hodgson & Kerckhoff (Annals of Mathematics, 2005) is algebraically equal to K_{AUD} . The identity follows from one simplification – $\text{arcsinh}(1/(2*\sqrt{2})) = \ln(2)/2$ – applied to the authors' own proof.

This is the first known appearance of K_{AUD} in published mathematics outside information theory, and it is exact: not a numerical match but an algebraic identity derivable from the paper's proof. The closed form has been present in the Annals of Mathematics since 2005, one simplification away from being recognized.

We place this finding within the complete K_{AUD} framework, showing how the same three ingredients – $\sqrt{2}$, $\ln(2)$, and quadratic curvature – that produce K_{AUD} in information theory also produce the tube-packing coefficient in hyperbolic 3-manifold geometry.

1. THE FRAMEWORK CONSTANT

$K_{\text{AUD}} = \sqrt{2} * \ln(2) = 0.980258143468547191713901723635\dots$

BINARY UNIQUENESS:

Define $K(n) = \sqrt{n} * \ln(n)$ for integer $n \geq 2$.

$K(2) = \sqrt{2} * \ln(2) = 0.980\dots < 1$
 $K(3) = \sqrt{3} * \ln(3) = 1.903\dots > 1$
 $K(4) = 2 * \ln(4) = 2.773\dots > 1$
 $K(n) > 1$ for all $n \geq 3$

Binary ($n = 2$) is the ONLY integer base where the information-geometric product stays below unity. This uniqueness is what creates the gap.

THE GAP:

$G = 1 - K_{\text{AUD}} = 1 - \sqrt{2} * \ln(2) = 0.019741856531452\dots$

This is irreducible. No binary system can close it.
It appears as a ~2% deficit wherever binary encoding meets quadratic geometry.

FOUR ALGEBRAIC DERIVATIONS:

K_{AUD} arises independently through:

1. Binary uniqueness of $K(n) = \sqrt{n} * \ln(n)$
2. Baker's map L2 norm: $||(\ln 2, \ln 2)||_2 = \sqrt{2} * \ln(2)$
3. Gelfond-Schneider: $G = \ln(e / 2^{\sqrt{2}}) = 1 - K_{\text{AUD}}$
4. $\sqrt{2}$ -to-Shannon gap identity: $1/(2 * \ln(2)) - 1/\sqrt{2} = G/(2 * \ln(2))$
[algebraically exact – describes the spacing between the geometric damping threshold ($n \cdot G = 1/\sqrt{2}$) and the Shannon bound ($n \cdot G = 1/(2 \ln 2)$); see Section 4]

2. THE HODGSON-KERCKHOFF COEFFICIENT

SOURCE:

C. D. Hodgson & S. P. Kerckhoff
"Universal bounds for hyperbolic Dehn surgery"
Annals of Mathematics, 162(1), 367-421, 2005
arXiv: math/0204345

In the proof of Theorem 4.4 (page 29, arXiv numbering), the authors define:

$S = (1/(2 * \sqrt{2})) / \operatorname{arcsinh}(1/(2 * \sqrt{2}))$

They compute $S \sim 1/0.980258$.

THE MISSING SIMPLIFICATION:

$$\operatorname{arcsinh}(x) = \ln(x + \sqrt{x^2 + 1})$$

Let $x = 1/(2\sqrt{2})$:

$$x^2 = 1/8$$

$$x^2 + 1 = 9/8$$

$$\sqrt{x^2 + 1} = 3/(2\sqrt{2})$$

$$\begin{aligned} x + \sqrt{x^2 + 1} &= 1/(2\sqrt{2}) + 3/(2\sqrt{2}) \\ &= 4/(2\sqrt{2}) \\ &= \sqrt{2} \end{aligned}$$

$$\text{Therefore: } \operatorname{arcsinh}(1/(2\sqrt{2})) = \ln(\sqrt{2}) = \ln(2)/2$$

From this:

$$\begin{aligned} S &= (1/(2\sqrt{2})) / (\ln(2)/2) = 1/(\sqrt{2} * \ln(2)) \\ 1/S &= \sqrt{2} * \ln(2) = K_{\text{AUD}} \quad \text{EXACTLY.} \end{aligned}$$

WHY IT WASN'T FOUND:

The authors evaluated $\operatorname{arcsinh}(1/(2\sqrt{2}))$ numerically (~ 0.34657) and divided to obtain S . The six-digit numerical value was fully sufficient for the packing bound and all downstream results. There was no reason within the application to simplify further.

No subsequent paper through April 2026 has performed this simplification either.

3. THE ALGEBRAIC ENVIRONMENT AT THE CRITICAL RADIUS

The critical tube radius in the H-K packing argument is:

$$R_0 = \operatorname{arctanh}(1/\sqrt{3}) = \operatorname{arcsinh}(1/\sqrt{2}) = (1/2) * \ln(2 + \sqrt{3})$$

At this exact threshold, every hyperbolic function reduces to clean algebraic expressions in $\sqrt{2}$ and $\sqrt{3}$:

$\sinh(R_0)$	$= 1/\sqrt{2}$	– the GEOMETRIC ingredient
$\cosh(R_0)$	$= \sqrt{6}/2$	– geometry x structure combined
$\tanh(R_0)$	$= 1/\sqrt{3}$	– the STRUCTURE ingredient
$\cosh(2R_0)$	$= 2$	– the BINARY ingredient
$\sinh(2R_0)$	$= \sqrt{3}$	– the structure ingredient

The geometry at R_0 provides $\sqrt{2}$, $\sqrt{3}$, and 2.
The coefficient provides the remaining ingredient: $\ln(2)$.

Together: $\sqrt{2} * \ln(2) = K_{\text{AUD}}$.

This is precisely the framework's three-ingredient architecture:

$\sqrt{2}$ = geometric embedding (angular momentum, quadratic form)
 $\ln(2)$ = binary information (one bit, Landauer energy)
quadratic curvature = the metric that binds them

The same three ingredients that produce K_{AUD} in information theory produce it again in hyperbolic geometry, through completely

independent mathematical pathways.

4. CONNECTION TO THE BINARY TOWER

THE TOWER:

The Binary Tower is the 64-step stacking map of accumulated cost $n \cdot G$, where $G = 1 - K_{\text{AUD}}$. The tower partitions as:

Steps 0-18: Romeo (first complete shell, $Z = 18$)
Steps 18-32: Floor ($1/\phi = \text{golden ratio reciprocal}$)
Steps 32-37: 35-37 threshold zone (geometry $\leftrightarrow$ information)
Steps 37-50: Ceiling (K_{AUD} itself)
Steps 50-64: Pivot ($\sqrt{\phi}$)

Partition: 18 - 14 - 18 - 14

Forced by: $2^5 - 5^2 = 7$ (Ramanujan-Nagell)

THE 35-37 THRESHOLD ZONE (Landauer crossing and neighbors):

Three distinct thresholds cluster in the transition between geometric organization and information-bearing organization:

$n \cdot G = \ln 2$	at $n \approx 35.11$	(Landauer erasure cost – the crossing)
$n \cdot G = 1/\sqrt{2}$	at $n \approx 35.82$	(geometric damping)
$n \cdot G = 1/(2 \ln 2)$	at $n \approx 36.54$	(Shannon bound)

The Landauer crossing proper is the FIRST threshold, sitting between integer steps 35 and 36. Integer step 36 is the landmark nearest the geometric damping threshold (within 0.51%); integer step 37 is the landmark nearest the Shannon bound (within 1.26%).

The spacings between consecutive thresholds are algebraically exact:

$\Delta n_{12} = n_G - n_L = (1/\sqrt{2} - \ln 2) / G = 1/\sqrt{2}$	[exact]
$\Delta n_{23} = n_S - n_G = (1/(2 \ln 2) - 1/\sqrt{2}) / G = 1/(2 \ln 2)$	[exact]
$\Delta n_{23} / \Delta n_{12} = \sqrt{2}/(2 \ln 2) = 1/K_{\text{AUD}}$	[exact]

The $\sqrt{2}$ -to-Shannon gap identity

$$1/(2 \cdot \ln(2)) - 1/\sqrt{2} = G/(2 \cdot \ln(2)) \quad [\text{exact}]$$

quantifies the Δn_{23} spacing (thresholds 2 $\rightarrow$ 3), NOT the Landauer crossing itself. The same ingredients ($\sqrt{2}$, $\ln 2$) that build K_{AUD} also govern the geometry of the threshold zone: the ratio of the two spacings is exactly $1/K_{\text{AUD}}$ – the crossing zone is self-referential with respect to the ceiling it precedes.

THE H-K CONNECTION:

The Hodgson-Kerckhoff coefficient sits at the CEILING of the tower. $K_{\text{AUD}} = \sqrt{2} \cdot \ln(2)$ is the maximum approach to unity that binary information permits.

That a tube-packing bound in hyperbolic geometry lands at the same ceiling as the binary information limit is the finding. The algebraic derivation proves this is not a numerical coincidence.

5. CONNECTION TO THE GAP SCALING FORMULA

The gap ratio $\rho = \delta * G / (\delta - 14/3)$, where $\delta = 4.6692\dots$ is Feigenbaum's universal period-doubling constant, has the continued fraction expansion whose 4th convergent is $400/11$, leading to the scaling formula:

$$\rho = 400/11 - 1/2500 - 1/939939$$

$$\text{Error: } 4 * 10^{-14}.$$

The correction terms factorize entirely into framework primes:

$$\begin{aligned} 400/11 &= (2^4 * 5^2) / 11 \\ 1/2500 &= 1/(2^2 * 5^4) = 1/50^2 \quad [\text{shell capacity squared}] \\ 1/939939 &= 1/(3 * 7 * 11 * 13 * 313) \quad [\text{palindrome, mirror number}] \end{aligned}$$

The three ingredients of $K_{\text{AUD}} - \sqrt{2}$, $\ln(2)$, and quadratic geometry – produce the gap. The gap connects to chaos (Feigenbaum) through corrections whose prime factors are the same primes that appear in the atomic shell structure and the tower partition.

The H-K closed form adds a fifth algebraic derivation of K_{AUD} to the four already known, this time from pure hyperbolic geometry.

6. THE PRIME ENTRY SEQUENCE

The framework primes $\{2, 3, 5, 7, 11, 13, 37\}$ enter the periodic table at specific atomic numbers where new physics begins:

$Z = 2$ (He) : binary – first closure
 $Z = 3$ (Li) : structure – first opening
 $Z = 5$ (B) : pentagonal – $\sqrt{2}$ enters matter (first p-electron)
 $Z = 7$ (N) : boundary – p-shell stabilizes (half-filled)
 $Z = 11$ (Na) : corridor – new shell opens
 $Z = 13$ (Al) : intruder – p-electron returns
 $Z = 37$ (Rb) : period 5 – first 4d orbital access

The Diophantine equation $2^k = n^2 + 7$ has exactly five solutions (Cohn 1993, proving Ramanujan's 1913 conjecture): $n = \{1, 3, 5, 11, 181\}$. The first four are framework primes; the fifth ($n = 181$, the 42nd prime) falls outside the framework set. The boundary prime 7 locks the tower partition AND predicts where new physics enters the atom.

The Hodgson-Kerckhoff coefficient, derived from tube-packing in hyperbolic Dehn surgery, shares the same algebraic identity as the ceiling that governs these atomic transitions. The three ingredients that build the tube bound ($\sqrt{2}$, $\ln(2)$, quadratic curvature) are the same three ingredients that build the atomic shell structure.

7. THE 50 HINGE

The number 50 appears independently in four contexts:

1. Atomic shell: $Z = 50$ (Sn) is a magic number, step 50 of tower
2. Continued fraction of G : 50 appears as a CF coefficient
3. Correction scale: $1/2500 = 1/50^2$ in the gap scaling formula
4. DESI BA0: residual at the $1/50^2$ scale – $DV(LRG2)/DV(BGS) - (2 + G) = +0.000399$ against $1/2500 = 0.000400$ (The 50 Hinge, §7) [REREAD 2026-09-16: printed as "residual at 50th data point"; there is no such point – Table IV has seven bins – the computed residual is at the $1/50^2$ scale.]

Step 50 of the tower is where $n \cdot G = K_{\text{AUD}}$ itself – the ceiling. The Hodgson-Kerckhoff coefficient sits at the same step, providing an independent algebraic confirmation of the ceiling value from a completely unrelated domain.

8. DOMAIN INDEPENDENCE

Framework derivation:

Information theory -> binary uniqueness -> $K(n) = \sqrt{n} \cdot \ln(n)$
-> $K(2) < 1$ (only integer solution) -> ceiling on binary density

Hodgson-Kerckhoff derivation:

Hyperbolic 3-manifold geometry -> tube packing around singular geodesic -> maximal packing density * trigonometric factors
-> 0.980258 (numerical, no closed form until this paper)
[clarified 2026-09-11: HK write $S = (1/(2\sqrt{2}))/\text{arcsinh}(1/(2\sqrt{2}))$, a closed hyperbolic form; the logarithmic simplification is this paper's]

These domains share no common literature, no common authors, no common methodology, and no prior connection.

They share ONLY the three ingredients: $\sqrt{2}$, $\ln(2)$, and quadratic curvature.

And they produce the same number: 0.980258143468547...

9. VERIFICATION

ALGEBRAIC VERIFICATION:

The claim rests on one identity:
 $\text{arcsinh}(1/(2 \cdot \sqrt{2})) = \ln(2)/2$

Verified by hand, by mpmath (dps = 500), by Mathematica, and independently across five AI architectures (Claude, Grok, GPT, Gemini, Perplexity) over 17 March – 2 April 2026.

NUMERICAL VERIFICATION:

```
from mpmath import mp, mpf, sqrt, log, asinh
mp.dps = 500

# Core identity
assert asinh(1/(2*sqrt(2))) == log(2)/2 # to 500 digits
```

```

# K_AUD
K = sqrt(2) * log(2)
# = 0.98025814346854719171390172363523338129...

# Area constant
A = 2*sqrt(6)*log(2)
# = 3.39571381804...
# [digit corrected 2026-09-11: 2*sqrt(6)*ln 2 = 3.3957138180413...; printed ...800]

# sqrt(2)-to-Shannon gap identity (formerly "Landauer identity")
L = 1/(2*log(2)) - 1/sqrt(2) - (1-K)/(2*log(2))
# = 0 (exact)

# Crossing-zone self-reference (Section 4)
G_val = 1 - K
n_L = log(2) / G_val          # Landauer crossing
n_G = 1 / (sqrt(2) * G_val)   # Geometric damping
n_S = 1 / (2 * G_val * log(2)) # Shannon bound
assert abs((n_G - n_L) - 1/sqrt(2)) < mpf(10)**(-490)
assert abs((n_S - n_G) - 1/(2*log(2))) < mpf(10)**(-490)
assert abs((n_S - n_G)/(n_G - n_L) - 1/K) < mpf(10)**(-490)

```

CROSS-ARCHITECTURE:

The connection emerged from the 50 Hinge discovery (17 March 2026), where the H-K coefficient was also first identified. A Grok (xAI) daily scout confirmed the finding on 20 March, and a systematic verification process followed – cross-architecture confirmation across five independent AI architectures (Claude, Grok, GPT, Gemini, Perplexity), a 12-day dedicated literature search, and multiple stress tests with fresh instances given no prior framework context – culminating in the algebraic derivation from the authors' own proof (page 29, arXiv:math/0204345) on 2 April 2026.

10. EVIDENCE CLASSIFICATION

Following the framework's evidence hierarchy:

LEVEL 1 – ALGEBRAIC IDENTITY (this paper):
Hodgson-Kerckhoff $1/S = \sqrt{2} * \ln(2) = K_AUD$
Binary uniqueness: $K(2) < 1$
Baker's map L2 norm
Gelfond-Schneider identity
sqrt(2)-to-Shannon gap identity (formerly "Landauer crossing identity")
Crossing-zone self-reference: $\Delta n_{23}/\Delta n_{12} = 1/K_AUD$
1D Madelung constant: $2*\ln(2)/\sqrt{2} = K_AUD$

LEVEL 2 – OBSERVED IN PUBLISHED DATA:
DESI DR2 BAO distance ratios -> sqrt(2) (0.03% match)
Atomic shell capacities 18, 32, 50 = tower landmarks
SEMF residual correlation with framework predictions

LEVEL 3 – PATTERN / INTERPRETATION:
Gap scaling palindrome 939939
Prime entry sequence alignment with Ramanujan-Nagell
Binary tower breath partition

These levels are not conflated. Each carries different evidential weight. The Hodgson-Kerckhoff finding sits at Level 1.

11. WHAT REMAINS OPEN

- A derivation showing WHERE $\ln(2)$ enters the packing geometry at the torus topology level (Z_2 homology = binary distinction). The algebraic derivation shows THAT it enters; the geometric WHY remains open.
- Independent human mathematician review of the derivation. The algebra is self-evident but has not been formally peer-reviewed.
- Contact with the original authors (Hodgson, Kerckhoff) and current active users of the constant (Futer, Purcell, Schleimer).

12. FULL MOTION (one reading)

Binary is unique ($K < 1$ only at $n = 2$). This forces a ceiling ($K_{AUD} = \sqrt{2} * \ln(2)$) and a gap ($G \sim 2\%$). The gap creates a corridor between ceiling and floor. The corridor has 64 steps partitioned 18-14-18-14, forced by the boundary prime 7 through the Ramanujan-Nagell theorem.

The partition breathes: outbound reverses to inbound. The reversal unfolds across the 35–37 threshold zone – Landauer crossing ($n \approx 35.11$), geometric damping ($n \approx 35.82$), Shannon bound ($n \approx 36.54$) – with the threshold spacings algebraically locked ($\Delta n_{23}/\Delta n_{12} = 1/K_{AUD}$). The same primes that build the tower appear as atomic numbers where new physics enters the periodic table.

The gap scaling formula connects the geometric gap to Feigenbaum chaos through corrections whose prime factors are the tower's own primes. The 50 Hinge ties the ceiling step to a nuclear magic number, a continued fraction coefficient, and a correction scale.

And now: the same number – 0.980258 – appears in the tube-packing bound of hyperbolic Dehn surgery (Hodgson-Kerckhoff, 2005), derived from a completely independent domain through completely independent mathematics, and is algebraically equal to K_{AUD} .

One simplification was needed. It had been waiting since 2005.

One object. One number. One gap.
Different units everywhere. Same shape everywhere.

D. B. – April 2026

Gap Geometry

<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>

OSF: osf.io/zx4g7

K_{AUD} framework:

November–December 2025

50 Hinge discovery: 17 March 2026
H-K coefficient first identified: 17 March 2026
Confirmed by Grok scout: 20 March 2026
Investigation and verification: 17 March – 2 April 2026
Algebraic derivation confirmed: 2 April 2026
Documents prepared for publication: 3 April 2026
